Learning the Opportunity Cost of Shared Drone Infrastructure: A Capacity-Monotone Reservation Model and Public-Data Replay Study

Research draft — authorship and affiliations to be supplied by the researchers

22 October 2026 (development record updated)

Status. TRC-oriented exploratory research draft. This document reports a completed computational pilot, including unfavorable comparisons. Novelty and submission readiness have not been established.

Abstract

Shared takeoff, landing, and charging resources couple current drone dispatch decisions to future delivery opportunities. We formulate an online single-depot reservation problem in which a dispatcher assigns each revealed parcel either to an unlimited ground fallback or to a feasible drone departure slot. A calendar records simultaneous requirements for pad access, fleet occupancy, and charging. We establish a resource-monotonicity property under an exogenous demand law and implement a temporal-difference learning policy with a capacity-monotone post-decision value approximation. A matched signed-weight model examines the effect of this structural restriction. The experiment replays publicly available LaDe-D Jilin task acceptance times and destinations under explicitly simulated drone settings. Chronological partitions contain 4,667 training, 2,559 validation, and 911 test tasks. Across 17 nonempty test days and five training seeds, the monotone model attains 69.05 generalized saving units per day, compared with 63.93 for greedy dispatch, 68.44 for its signed-weight counterpart, and 74.93 for a validation-tuned resource-price policy. Thus, the pilot does not establish a performance advantage over the strongest implemented online baseline. A reduced-fleet stress test further exposes weak transfer. The study provides a reproducible reservation benchmark and a falsifiable structural-learning hypothesis; it does not establish deployment benefits or state-of-the-art performance.

Keywords: drone logistics; online scheduling; shared infrastructure; reinforcement learning; approximate dynamic programming; reproducibility.

1 Introduction

An available aircraft does not by itself make a drone delivery feasible. Its departure requires access to a launch facility, its return requires landing capacity, and its next assignment may require a charging position. A short trip can therefore prevent a more valuable later trip even when its immediate saving is positive. This motivates a dispatch question: how much future infrastructure capacity should an operator preserve when the next delivery requests are unknown?

We study a deliberately restricted version of this question. All sorties start and end at one depot, aircraft are identical, travel times are deterministic, and orders are served individually. A departure decision reserves the entire sortie and charging interval. Ground fallback is always available at a modeled reference cost. These assumptions remove routing and repositioning choices so that the experiment can isolate intertemporal competition for shared resources. The objective is expected saving relative to the ground reference, rather than a service-rate increase: with unlimited ground fallback, the model cannot demonstrate improved overall fulfillment.

The proposed experiment asks whether a structural restriction on the learned continuation value improves online reservation decisions. More remaining capacity cannot worsen the optimal continuation value when extra capacity can be left unused. We impose this monotonicity directly in a compact neural approximation and compare it with an otherwise matched approximation permitting signed capacity weights. The resulting policy can be interpreted as balancing immediate saving against a learned opportunity cost of the reservation.

Three outputs are developed. First, a time-indexed resource model connects online decisions to a perfect-information integer-programming benchmark. Second, a simple feasible-policy inclusion argument motivates the monotone approximation while clearly separating the proved property from additional architectural assumptions. Third, a public-data replay protocol distinguishes observed delivery fields from simulated drone operations and preserves unfavorable results. These are concrete methodological outputs, not evidence that the combination is globally novel. In particular, the experiment finds that a simple tuned resource price outperforms the learned policies. That finding defines the boundary of the present contribution and a useful target for subsequent work.

2 Related work and positioning

Chen, Ulmer, and Thomas [1] study dynamic same-day service with vehicles and drones using deep Q-learning, including the value of state and action information. Campuzano, Lalla-Ruiz, and Mes [2] formulate dynamic drone scheduling at a central station and use value-based reinforcement learning. Consequently, depot-based stochastic drone dispatch and anticipatory value learning are established ideas.

Hu, Du, and Li [3] address a richer multimodal food-delivery system in which couriers bring parcels to launchpads and drones complete subsequent legs. Their reported framework combines multi-agent learning, risk-aware assignment, and coordination across transfers. Yu, Yu, and Wu [4] report feasibility masks for taxi–drone cooperation, including recovery and docking constraints. Neither learning-based cooperation nor masking resource infeasibility can serve as the novelty claim here. The current pilot uses a simpler centralized reservation system and does not reproduce those algorithms or compare their numerical performance across incompatible simulators.

Structural value approximation also has a substantial history. Jiang and Powell [5] develop monotone approximate dynamic programming. Asadi and Nurre Pinkley [6] exploit monotonicity for battery-swap operations relevant to drones and electric vehicles. We do not introduce monotone ADP or transfer their convergence guarantees to the neural temporal-difference procedure used below. The candidate distinction is the explicit joint calendar of pad, sortie, and charging occupancy and the controlled assessment of an increasing calendar-value approximation. Establishing whether that distinction is sufficient for a journal contribution requires a broader full-text comparison.

The literature checks supporting this draft cover publisher or institutional metadata, abstracts, and accessible excerpts; they are not a systematic full-text review. Accordingly, absence of an identical formulation in the reviewed sources is not evidence of priority. The public LaDe dataset [7] supplies a reproducible demand trace, but its courier task events are not measurements of drone operations.

3 Operational model

3.1 Information and assumptions

An episode comprises a finite sequence of revealed orders $i = 1, \ldots, N$. Each order has a reveal slot t_i, depot-to-destination distance d_i, and an assumed delivery allowance D_i. At its

decision epoch, the operator observes that order, previous revealed orders and decisions, and outstanding reservations. Future orders and the episode's eventual order count are not policy inputs. Orders sharing a slot are disclosed sequentially in a fixed hashed-identifier order; this artificial convention is shared by every policy.

Let $\mathcal{T} = \{0, \dots, T-1\}$ be five-minute slots and $\mathcal{R} = \{P, F, C\}$ denote pad, fleet, and charger resources. Capacities are $B_P = 1$, $B_F = 4$, and $B_C = 2$ in the base experiment. The horizon has $T = 60$ slots, allowing arrivals during its first 48 slots and time for returns thereafter. Every drone returns to the same depot and completes a fixed two-slot charging operation before becoming available. The model does not track battery state, payload, airspace conflicts, weather, or charging power. The six-kilometer one-way eligibility limit is an assumed range screen, not an energy-feasibility calculation.

Ground fallback has no resource or deadline constraint in this pilot. Its cost is a reference generalized effort, not an observed courier cost. Every order is therefore assigned, and all-aircraft feasibility statements refer only to the modeled drone reservations. They imply neither operational flight safety nor empirical delivery reliability.

3.2 Reservation patterns and costs

For an eligible order, action $a \in \{1, 2, 3, 4\}$ denotes a departure delay $\delta = a - 1$ slots. Action zero assigns ground fallback. Set

$$f_i = \max\{1, \lceil d_i/(v_D\Delta)\rceil\}, \qquad \ell_{ia} = t_i + \delta, \tag{1}$$

$$b_{ia} = \ell_{ia} + 2f_i + 1, \qquad e_{ia} = b_{ia} + q, \tag{2}$$

where $v_D = 0.8$ km/min, $\Delta = 5$ min, and $q = 2$. Here ℓ, b, and e denote launch, landing, and ready-for-reuse slots. The extra return slot conservatively accommodates handling. Customer delivery time relative to reveal is $\delta\Delta + f_i\Delta + 2$ minutes. A candidate must satisfy $d_i \le 6$, $\delta\Delta + f_i\Delta + 2 \le D_i = 30$, and $e_{ia} < T$. The strict horizon cutoff is part of this benchmark.

Each action has a nonnegative integer resource pattern $u_{ia,r\tau}$:

$$u_{ia,P\tau} = \mathbf{1}\{\tau = \ell_{ia}\} + \mathbf{1}\{\tau = b_{ia}\}, \tag{3}$$

$$u_{ia,F\tau} = \mathbf{1}\{\ell_{ia} \le \tau < e_{ia}\}, \tag{4}$$

$$u_{ia,C\tau} = \mathbf{1}\{b_{ia} \le \tau < e_{ia}\}. \tag{5}$$

Charging and the landing slot overlap by convention. The pad is not occupied during other charging slots. Aircraft are indistinguishable; interval occupancy with a common depot permits a consistent aircraft assignment whenever the fleet bound holds. This argument would fail for heterogeneous aircraft or different return depots without additional state.

The reference and drone costs are

$$c_i^G = 2\rho d_i/v_G + 4, \qquad c_{ia}^D = 2d_i/v_D + 4 + 0.15\delta\Delta, \tag{6}$$

with road factor $\rho = 1.35$ and $v_G = 0.35$ km/min. Saving is $r_{ia} = c_i^G - c_{ia}^D$; ground has $r_{i0} = 0$ and zero resource consumption. These dimensionless generalized units combine travel effort and a waiting penalty. They are not monetary, emissions, or welfare estimates. Equal handling constants cancel in the saving objective.

3.3 Online decisions and offline benchmark

Let C_i be the remaining resource calendar before order i. The feasible action set contains ground and all physically eligible patterns satisfying $u_{ia} \le C_i$ componentwise. On selecting a_i,

the calendar becomes $C_i - u_{ia_i}$. Reservations cannot be canceled. For a fixed finite demand distribution, the online objective is

$$\max_{\pi \text{ nonanticipative}} \mathbb{E}_\pi \left[\sum_{i=1}^{N} r_{ia_i} \right]. \tag{7}$$

Nonanticipativity requires identical decisions for histories indistinguishable to the operator; it excludes use of future test requests.

For a realized full day, a binary pattern program gives the perfect-information benchmark:

$$\max_x \quad \sum_i \sum_{a \in A_i^D} r_{ia} x_{ia} \tag{8}$$

$$\text{s.t.} \quad \sum_{a \in A_i^D} x_{ia} \le 1 \qquad \forall i, \tag{9}$$

$$\sum_i \sum_{a \in A_i^D} u_{ia,r\tau} x_{ia} \le B_r \qquad \forall r, \tau, \tag{10}$$

$$x_{ia} \in \{0, 1\}.$$

The implicit unselected option is ground service. A_i^D includes physical candidate patterns generated without prior reservations. Knowing the whole day makes this benchmark more informed than every online policy. Its optimal objective bounds attainable savings from above under the same patterns. With solver tolerances or a time limit, the reported dual bound, rather than an arbitrary incumbent, is the certified upper bound. We retain both and the termination status.

4 Capacity-monotone post-decision learning

4.1 Structural property and its limits

Write h for fixed revealed information determining the conditional law of future demand. Let $W(h, C)$ be the optimal expected continuation saving after the current decision, with remaining calendar C.

Proposition 1. *If future demand is exogenous to decisions, unused capacity has no penalty, and ground fallback remains feasible, then $C' \ge C$ implies $W(h, C') \ge W(h, C)$ for the same h and conditional demand law.*

Proof. Take any feasible nonanticipative policy for calendar C. Under C', maintain a virtual copy initialized at C and make the decisions prescribed by that policy from its virtual history. Couple both systems to the same future requests. Each selected pattern is feasible in the virtual system. The actual calendar exceeds the virtual calendar by $C' - C$ after every identical subtraction, so that pattern remains feasible in the actual system. Both systems obtain the same saving on every coupled path. Thus every attainable expected saving under C is attainable under C'. Taking the supremum over policies proves the result. □

The proposition does not establish concavity, separability, or monotonicity when the demand law changes with capacity. Nor does it prove that time and a calendar alone are a sufficient Markov state for the observed demand process. In the implementation, historical information is compressed, so the value network is a heuristic partial-information approximation.

4.2 Architecture and opportunity cost

Let z contain normalized remaining capacities, with entries for past slots set to zero. Temporal features are $\eta(t) = (t/T, \sin(\pi t/T), \cos(\pi t/T), 1)$. The implemented approximation is

$$\widehat{W}_\theta(t, z) = b_\theta(\eta(t)) + \sum_{j=1}^{3T} \text{softplus}(g_{\theta,j}(\eta(t)))\,[1 - \exp(-z_j)]. \tag{11}$$

At fixed time, every partial derivative with respect to z_j is nonnegative. The capacity transformation also imposes a separable diminishing-return shape; this is an architectural assumption that is stronger than the proposition and may be wrong for complementary resources. The temporal weight network has one 32-unit tanh hidden layer, and the intercept has one 16-unit tanh hidden layer. The signed-weight ablation replaces softplus weights with unrestricted weights and uses matching initial effective weights. Different parameterizations still induce different optimization dynamics, so the comparison does not isolate monotonicity perfectly.

At each order the policy selects

$$a_i \in \arg\max_{a \in A_i(C_i)} \{r_{ia}/20 + \widehat{W}_\theta(t_i, z(C_i - u_{ia}))\}. \tag{12}$$

Subtracting the action-independent value at C_i shows the implied opportunity cost of consuming u_{ia}. Feasibility is enforced by enumerating admissible patterns, independently of learned values. Ground fallback therefore remains selectable even when predictions are poor.

4.3 Learning procedure

The value describes rewards *after* the current action. For a stored selected afterstate z_i, the temporal-difference target begins with the next revealed order's reward:

$$a^* \in \arg\max_{a \in A_{i+1}} \{r_{i+1,a}/20 + \widehat{W}_\theta(z_{i+1,a})\}, \tag{13}$$

$$y_i = r_{i+1,a^*}/20 + \widehat{W}_{\bar{\theta}}(z_{i+1,a^*}). \tag{14}$$

At episode termination $y_i = 0$. Adding the current reward again would be an indexing error. We use an online network for target action selection and a periodically copied network for target evaluation, a replay buffer of 50,000 transitions, batches of 64, Huber loss, Adam with learning rate 0.001, gradient norm limit 5, one update per four decisions, and a target copy every 100 updates. The finite-horizon discount is one. Exploration decreases from 0.7 to a floor of 0.05 during the first 80% of 400 sampled training days. Each day is sampled uniformly with replacement from the training partition. A validation check every 50 episodes retains the best checkpoint. No convergence or optimality guarantee is asserted for this procedure.

5 Public-data replay and evaluation

5.1 Data construction

The official LaDe-D Jilin file contains 31,415 records across 163 distinct day identifiers. We sort those identifiers, allocate the first 60% to training, the next 20% to validation, and the remainder to testing before selecting a region or estimating a depot. The most frequent training region is 53. The artificial depot is the componentwise median of valid training acceptance GPS coordinates in that region: longitude 126.56689, latitude 43.81213. This point is not a verified drone facility.

Within each partition, we retain tasks accepted between 07:00 and 11:00 and compute approximate planar distances to the depot. Acceptance timestamps are rounded down to five-minute slots; the simulator reveals tasks at those discretized times. This is a new discretized

replay process, not exact real-time service at the original timestamps. The CSV's courier acceptance event must not be interpreted as customer order placement. The selected file does not specify a calendar year; day identifiers remain in month–day form. The complete input SHA256 and the transformation are included in the repository.

Table 1: Nonempty replay partitions after filtering.

Partition	Days	Tasks	First–last retained day
Training	89	4,667	05-14–08-18
Validation	33	2,559	08-19–09-20
Test	17	911	09-21–10-21

Only destinations and courier acceptance events are empirical inputs. Drone capacities, speeds, deadlines, range, charging times, handling, and costs are simulated assumptions. The official dataset card identifies an Apache-2.0 license. We distribute a downloader and checksums rather than republishing order records. Five test days contain only one retained task. These days remain in the primary evaluation, and the resulting 17-day sample is too small for strong generalization claims.

5.2 Baselines and statistical protocol

Ground-only service gives zero reference saving. Greedy maximizes current saving over feasible candidates. A resource-price policy maximizes

$$r_{ia} - p(1 - t_i/T) \sum_{r,\tau} u_{ia,r\tau} / \max(B_r, 1). \tag{15}$$

The price is selected on validation days from $\{0, 1, 2, 4, 8, 12, 20, 32\}$; the selected value is 4. Sampled rollout draws three training days at each decision, takes their orders strictly after the current slot as possible futures, and evaluates greedy continuations from each candidate after-state. It uses no actual test future. Its unconditional day sampling and omission of unrevealed same-slot tasks make it a limited scenario baseline, not an optimized stochastic controller.

Both learning variants use seeds 11, 22, 33, 44, and 55 and 400 training episodes per seed. Policies and parameters are frozen before test evaluation. No revision is selected using test performance. The MILP uses a 30-second time limit and relative gap tolerance 10^{-4}; all 17 base-test instances returned successful status. Their incumbents are reported separately from stored dual bounds.

For comparisons we average the five learned-policy outcomes within each day and then form paired day differences. The 85 seed–day observations are not treated as 85 independent demand realizations. Ten thousand fixed-seed bootstrap replicates provide descriptive percentile intervals. We also report circular blocks of three consecutive *observed* days because serial dependence may invalidate independent-day resampling. These intervals condition on the five fitted models, and do not measure the full uncertainty from retraining, model selection, or a different city.

6 Results

The monotone model improves mean saving over greedy by 5.12 units (8.01%), but falls below resource pricing by 5.88 units (7.85%). Its mean remains approximately 28.9% below the perfect-information incumbent, which has an informational advantage. Across training seeds, monotone daily means range from 67.82 to 71.10, while signed-weight means range from 65.66 to 70.00. All evaluated online episodes satisfy the simulator's resource checks. This outcome is enforced by candidate feasibility and cannot be attributed to learning a safety mechanism.

Table 2: Held-out performance. Higher saving is better. Decision time is the unweighted mean of daily means; ground is an unlimited reference.

Policy	Saving/day	Drone tasks/day	Decision ms
Ground	0.00	0.00	0.208
Greedy	63.93	5.41	0.147
Sampled rollout	65.93	5.41	31.628
Resource price	74.93	5.00	0.154
Signed-weight RL	68.44	4.80	0.424
Monotone RL	69.05	4.65	0.437
Perfect-information MILP incumbent	97.11	—	—

Table 3: Monotone RL minus comparator: paired daily saving differences and descriptive 95% percentile intervals.

Comparator	Mean difference	Independent-day interval	3-day block interval
Greedy	5.12	[0.08,10.44]	[-0.56,11.09]
Resource price	-5.88	[-10.04,-1.92]	[-9.27,-2.67]
Sampled rollout	3.12	[-1.84,8.54]	[-2.12,8.32]
Signed-weight RL	0.61	[-1.98,3.34]	[-1.70,3.62]

The independent-day interval against greedy only narrowly excludes zero; the block interval includes it. Both intervals for the architecture comparison include zero. The study therefore does not establish that monotonicity materially improves performance, and it should not be summarized as a significant superiority result. By contrast, the observed shortfall against resource pricing is consistent across the reported resampling schemes. These are exploratory comparisons without a multiple-testing correction.

Table 4: Frozen-policy fleet sensitivity: saving/day. Pad and charger capacities remain unchanged.

Policy	Two aircraft	Four aircraft	Six aircraft
Greedy	49.76	63.93	63.93
Resource price	59.06	74.93	73.90
Signed-weight RL	48.68	68.44	67.45
Monotone RL	47.73	69.05	68.79

The reduced-fleet case is particularly unfavorable to the learned policies. Monotone RL achieves less saving than greedy with two aircraft. Increasing the fleet to six does not improve these online outcomes, although the optimal value cannot decrease. This is not a contradiction: policy quality may deteriorate, and normalization divides each calendar by its scenario capacity, so the monotonicity guarantee applies within a fixed representation and context rather than across these differently normalized scenarios. An initially full fleet looks identical to the network regardless of its absolute size. Transfer to changed capacities is consequently a substantial unresolved issue.

A post hoc diagnostic excluding the five singleton days leaves 12 days. Their mean savings are 91.25 for monotone RL, 90.22 for signed-weight RL, 99.12 for resource pricing, and 83.53 for greedy. The principal ordering persists. This diagnostic does not replace the prespecified full test sample. The complete run took approximately 520 seconds on the local CPU environment with one PyTorch computation thread; timings include training and evaluation and are descriptive,

not a hardware-independent speed claim.

7 Discussion and research agenda

The experiment supports a narrow operational lesson: reserving capacity using even a coarse opportunity price can improve on immediate-saving dispatch in this constructed system. It does not support preferring the implemented neural policy to a calibrated price. One plausible explanation is that the separable continuation approximation misses complementarities: a free pad slot has little value without a simultaneously available aircraft and charger. A second is that time-only demand context cannot recognize day-specific intensity or spatial patterns. A third is limited training and seasonal change across chronological splits. These are hypotheses; none is established by the current experiment.

A promising next methodological study would learn nonnegative interaction terms across aligned pad, fleet, and charger windows while conditioning on causal recent-demand summaries and absolute capacities. Exact small-instance continuation values could provide diagnostics or training targets. That would require fresh development partitions and a new untouched test period; the present held-out results must not become an informal tuning set. Stronger conditional stochastic lookahead and a carefully adapted published RL baseline are needed before a performance claim.

The proposed model also omits weather-dependent flight duration, uncertain charging completion, multi-depot repositioning, finite ground resources, parcel weight, and actual facility operating constraints. Adding all of them at once would obscure the contribution. A defensible next scenario is uncertain return-time reservations with an explicit probability or robust-feasibility requirement, supported by accessible operational data. Any claimed improvement would then require both a calibrated disturbance model and constraint-violation evaluation outside training. No such claim is made here.

8 Conclusion

We formulate online drone dispatch as a joint infrastructure-reservation problem and implement a capacity-monotone post-decision learning policy. The resource-monotonicity argument is valid under stated assumptions, while the neural approximation introduces additional restrictions that need empirical justification. A chronological public-data replay yields a reproducible pilot with an informative negative result: monotone learning exceeds greedy on average but loses to tuned resource pricing and shows weak reduced-fleet transfer. The formulation, checks, and open experiment are a basis for further research. A TRC submission would require a sharper verified distinction from prior work, stronger baseline comparisons, richer data validation, and evidence beyond this pilot.

Reproducibility and declarations

The repository contains the data downloader, input hash, transformation, simulator, training code, tests, daily results, training logs, and statistical summary. Python 3.14.3, NumPy 2.4.2, SciPy 1.17.1, and PyTorch 2.10.0+cpu were used; the machine-readable experiment records are authoritative for versions. All experiment numbers in this draft derive from saved outputs. The original LaDe records and the researchers' private literature library are excluded from the release. No real drone deployment or controlled operational experiment was conducted. Authorship, affiliations, funding, conflicts of interest, and final data-use declarations must be completed and verified by the human researchers before submission. Generative AI assisted formulation, code, analysis, and drafting; human authors must verify and take responsibility for

any submitted version. Current TRC formatting and declaration requirements have not been verified in this draft.

A Development extension: joint-scenario reservations and resource-pattern values

This appendix records an exploratory successor to the completed pilot above. It does not replace the original experiment or claim a confirmed modeling or algorithmic contribution. The original test outcomes have already been observed, and a new independent confirmation set has not been evaluated. The current aim is to distinguish an operational mechanism from a change of function approximator.

A.1 Positioning and information restriction

Targeted full-text inspection of Chen et al. [1], physical pages 5–11 of their preprint, confirms that accepted but unloaded requests may remain in updated tentative plans. This mechanism cannot be claimed as new here. Finite charging capacity already appears in joint aerial-vehicle routing and charging models [8], and energy-predictive drone-service planning already includes charging reservations [9]. NASA's discussion of uncertain arrival and resource availability motivates examining the cost of operational buffers [10]; it does not calibrate our disturbances or validate flight safety.

We instead examine a finite set Ω of *joint* operating scenarios, with positive probabilities q_ω summing to one. Each scenario specifies a common flight-time multiplier m_ω and an integer charging duration c_ω for the operating period. These are synthetic assumptions. A request is assigned either to the ground reference or to a launch slot $\ell_i = a_i + d_i$ fixed at acceptance. The launch decision is identical across scenarios. The current controller deliberately retains its original forecast set; it does not condition on subsequently observed flight outcomes. It is therefore a restricted observation policy, not an optimal controller using every available operational observation.

A.2 Scenario-dependent occupancy with common commitments

Let Δ be slot duration, v nominal drone speed, and s_i distance. For a candidate delay d, define

$$f_{i\omega} = \max\{1, \lceil m_\omega s_i/(v\Delta)\rceil\}, \qquad b_{id\omega} = a_i + d + 2f_{i\omega} + 1, \tag{16}$$

$$e_{id\omega} = b_{id\omega} + c_\omega. \tag{17}$$

The pad is occupied at launch and landing, the aircraft on $[a_i + d, e_{id\omega})$, and a charger on $[b_{id\omega}, e_{id\omega})$. Write this resource-time vector as $U_{id\omega}$. Each scenario must satisfy the delivery deadline $\Delta(d + f_{i\omega}) + 2 \le \delta_i$ and $e_{id\omega} \le T$. Unlike the original pilot's strict horizon convention, completion exactly at T is allowed in this extension. The convention is tested explicitly and results from the two formulations are not pooled.

For remaining scenario calendars C_ω, an action is admissible only if $U_{id\omega} \le C_\omega$ for every ω. Once accepted, its common launch commitment is retained. With unlimited ground fallback, induction over admissions proves resource and deadline feasibility for every listed scenario: the initial calendars are feasible, each accepted action subtracts no more than the remaining capacity in any coordinate, and fallback subtracts zero. Homogeneous aircraft at one base can be assigned at launch to an available aircraft; the interval load constraint ensures one exists in each realized listed scenario. The argument does not cover omitted disturbances, physical airspace separation, or a safety certification.

Expected immediate saving is $r_{id} = \sum_\omega q_\omega(g_i - c_{id\omega})$, with the pilot's continuous generalized transport cost scaled by m_ω and its waiting penalty retained. Occupancy uses rounded slots while

cost uses a continuous proxy; these are separate modeling choices. The corresponding perfect-future integer model has objective $\max \sum_{i,d} r_{id} x_{id}$, constraints $\sum_d x_{id} \le 1$ and $\sum_{i,d} U_{id\omega} x_{id} \le K$ for every scenario, and binary x. It is an informational upper benchmark, not an online competitor.

A per-request envelope instead reserves $\overline{U}_{id} = \max_\omega U_{id\omega}$ coordinatewise. Since $U_{id\omega} \le \overline{U}_{id}$, envelope-feasible selections are joint-scenario feasible. The inclusion can be strict, a standard robust-optimization observation rather than a new general theorem. In a two-request example, launches at slots 0 and 2 return at $(3, 5)$ under one common scenario and $(5, 7)$ under the other. Both schedules satisfy pad capacity one, whereas the per-request envelope double-books slot 5. Exhaustive enumeration with the specified synthetic costs gives 15.3571 for joint reservations and 7.6786 for the envelope. These numbers establish a structural difference, not an operational percentage improvement.

A.3 Pattern features and a falsifiable method hypothesis

For a fixed future-request template p and its launch alternatives d, define

$$B_p(C) = \max_d \min \left\{ 1, \min_{\omega,r,t:U_{pd\omega,rt}>0} \frac{C_{\omega,rt}}{U_{pd\omega,rt}} \right\}. \tag{18}$$

Templates with no physics-feasible alternative have value zero. They are fixed distance-time templates, not future evaluation requests. The proposed approximate continuation value is

$$\widehat{V}_\theta(C, h) = b_\theta(h) + \sum_p \mathrm{softplus}(w_{\theta p}(h)) B_p(C). \tag{19}$$

Each B_p and their positive weighted sum are nondecreasing in remaining capacity for fixed context. Taking a maximum over launch alternatives does not generally preserve concavity. Templates compete for resources, so their summed value is neither an exact continuation value nor a certified upper bound.

The construction can express a basic complementarity that the pilot's separable critic cannot. A single future task worth R requiring both binary resources p and f has values $(0, 0, 0, R)$ at states $(0, 0), (1, 0), (0, 1), (1, 1)$. Its mixed finite difference is R, while any separable $g(p)+h(f)+b$ has mixed difference zero. Uniform error is therefore at least $R/4$ by the triangle inequality; a single feature $R \min(p, f)$ represents all four values exactly. This is an expressivity diagnostic, not a proof of improved learning or a claim that resource complementarity is new.

An independent code review exposed an important information-resolution issue: current-slot requests are revealed sequentially, but the first prototype omitted all future templates in the current slot. In the validation data, 2,279 of 2,559 requests are not the first request of their slot. The revised features retain current-slot opportunities and include only observed within-slot progress, cumulative revealed demand, recent revealed demand and mean distance, plus time features. They never use the actual remaining request count. The resulting summary is still not proved to be a sufficient Markov state.

The accompanying baseline selects up to three training-day continuations conditional on observed demand and within-slot progress, then solves a scenario-capacity LP for each candidate afterstate. Current actions share the same predicted paths. These pathwise LP values relax integrality and know the sampled future, so the baseline is an optimistic two-stage heuristic. It is not allowed access to the actual evaluation suffix. Empirical-support misses and solver fallbacks are recorded. Development comparisons and their limitations are stored separately from the original experiment; independent confirmation, disturbance calibration, and a fuller novelty review remain required.

A.4 Development outcomes, not independent confirmation

The revised development run uses the 89 original training days and 33 validation days, with 200 sampled training episodes for each of five seeds (11, 22, 33, 44, 55). All three critics receive the same eight-dimensional observed context; their parameter counts and feature scales differ. Checkpoint selection and price/lookahead tuning use these validation days, so the following comparison is exploratory.

Table 5: Development validation: mean simulated saving per day.

Policy	Saving/day
Joint-scenario greedy	76.15
Separable calendar TD	77.21
Unrestricted MLP TD	78.19
Resource price	80.88
Resource-pattern TD	83.54
Conditional training-path LP	84.13

Resource-pattern TD improves over the selected price by 3.28% but remains 0.70% below the conditional LP. Its five seed means range from 82.47 to 85.06; the best seed is not substituted for the mean. All 693 stored policy-day evaluations, including alternative lookahead weights, pass the finite-scenario audit; they represent only 33 distinct demand days. The selected lookahead weight of 0.5 encounters four empirical count-support misses and no solver failures. These results motivate further investigation of interactions, but do not meet the intended superiority criterion or establish confirmed novelty. The source, checkpoints, daily outcomes and the earlier unsuccessful prototypes are retained separately. Newly acquired public single-drone measurements have not yet calibrated this model and cannot by themselves establish multi-aircraft disturbance dependence.

B Partial dependence, joint capacity risk, and a negative method diagnostic

The following experiments were completed on 4 October and checked against saved source identities on 22 October 2026. They revise the scope of the preceding development mechanism. They do not constitute independent confirmation, a validated disturbance model, or a demonstrated algorithmic contribution.

B.1 What the public flight measurements support

We audited all 209 logs from the public CMU quadcopter dataset [11], preserving ground and hover records separately. The parameter file identifies 196 records with full-flight route labels. Their median whole-log duration is 184 seconds, median horizontal velocity-integrated path is 458.06 metres, and median voltage–current-integrated energy is 20.73 Wh. The corresponding 5th–95th percentile intervals are 135.88–252.85 seconds, 445.01–480.94 metres, and 14.21–29.00 Wh. These quantities include preflight and postflight logging; no phase-segmented airborne duration is claimed.

No nonfinite measurements or nonincreasing timestamps were found by the stated checks. Whole-log path totals differ from the source paper's approximate distance summary; that discrepancy remains unresolved rather than being removed by rescaling. The data concern short controlled routes and contain neither charging-site measurements nor simultaneous multi-aircraft operations. They therefore do not calibrate our kilometre-scale missions, five-minute discretization, charging durations, or dependence parameters. Relative airflow measurements also require

correction before interpretation as ambient wind. These limitations prevent using the dataset as evidence for the earlier common-disturbance assumption.

B.2 A fixed-marginal perturbation invalidates the full-support guarantee

Let $Z \in \{0, 1\}$ have positive prior $q = (q_0, q_1)$. Conditional on Z, each accepted task has an independent label Y_i with

$$\Pr(Y_i = y \mid Z = z) = (1 - \eta)\mathbf{1}\{y = z\} + \eta q_y, \qquad 0 \le \eta \le 1. \tag{20}$$

Every task retains marginal distribution q, while the pairwise correlation of the binary labels is $(1 - \eta)^2$. This is a synthetic dependence stress test, not an estimated weather model. For any $\eta > 0$, every binary label vector has positive probability. Consequently, all-support robust feasibility is equivalent to

$$\sum_{i \in A} \max_{y \in \{0,1\}} U_{iy,rt} \le K_{rt} \quad \text{for every resource–time coordinate } (r, t). \tag{21}$$

Indeed, maximization of a coordinate's sum over the Cartesian product of task labels separates across tasks; requiring every coordinate for every vector gives the displayed inequalities. Thus the previous common-scenario feasible-set advantage over an individual envelope vanishes under any positive resampling probability when the requirement is robustness to every supported outcome. This elementary support argument is not claimed as a new general theorem.

For the two-task example in Appendix A, the unique conflicting label pair has probability $q_0 q_1 \eta(2 - \eta)$. At $q = (0.7, 0.3)$ and $\eta = 0.001$, correlation is still 0.998001 but the conflict probability is 0.00041979. At $\eta = 0.01$ and 0.05, it is respectively 0.004179 and 0.020475. Passing the two common-scenario checks therefore does not certify feasibility under partially independent perturbations.

B.3 Restricted joint-chance admission and finite-law evaluation

The successor candidate retains the common-scenario hard constraints and adds a joint constraint over the entire reservation set:

$$r(A) := \Pr\left(\exists (r, t) : \sum_{i \in A} U_{iY_i,rt} > K_{rt}\right) \le \alpha. \tag{22}$$

The same α applies to the union of all resource–time conflicts across the entire accepted set. It is not renewed independently at each admission or allocated separately to every coordinate. Launch commitments remain irrevocable, and task outcomes are never revealed to this restricted controller. Demand is assumed independent of the synthetic disturbance law; no observed weather is used to select favorable conditional probabilities. Keeping the common-scenario hard constraints is an additional conservative restriction, so the experiment does not explore the unrestricted joint-chance feasible set.

The probability oracle uses variable elimination along time. It introduces a task label at its first possible occupied slot, checks current loads, and retires it after its last possible occupied slot. It retains separate joint probability masses for the two values of Z when merging retired labels; averaging these masses earlier would discard cross-window dependence. The method has exponential worst-case dependence on the active label frontier, rather than a polynomial worst-case claim. A state-count or mask-width limit returns unresolved status with risk upper bound one, and the candidate is rejected. Outward rounding and positive-summation error bounds propagate a lower bound on success probability used for admission. "Exact" denotes evaluation of the specified finite law by variable elimination, with probability arithmetic controlled as stated, not a real-world reliability certificate.

Seven oracle tests and an independent review's 120 additional rational-enumeration cases found no false acceptance under that law. A review also exposed a diagnostic-summary failure when an unresolved probability was averaged as a number; regression tests now require explicit unresolved counts and an all-record risk-upper-bound mean. The original and repaired diagnostic outputs are retained separately. At $\eta = 0.01$, 39 of the earlier pattern policy's 165 seed–day records exceed a 1% joint-risk threshold without the new filter. These records cover only 33 distinct demand days. Greedy and resource pricing exceed the threshold on 4 and 3 of those days. This is evidence about omitted disturbances in the earlier model, not evidence of superiority for a new learning method.

B.4 Risk-insertion features and their first falsification

For unit occupancy $U_{py,rt} \in \{0, 1\}$, let L_{rt} denote the load from accepted set A. Conditional independence gives the following bound on additional joint failure:

$$r(A \cup \{p\}) - r(A) \le D(A, p), \tag{23}$$

$$D(A, p) := \sum_{z} q_z \sum_{r,t} \Pr(L_{rt} = K_{rt} \mid z) \Pr(U_{pY_p,rt} = 1 \mid z). \tag{24}$$

To see this, a newly failed set that was previously feasible must contain a coordinate that was exactly full and receives unit additional load. Apply the union bound over these events and conditional independence. A truncated Poisson-binomial recursion computes the saturation probabilities. The bound can count a single failing trajectory multiple times; it is not an exact marginal failure probability.

The candidate critic replaces each ordinary insertion feature by its core-feasibility indicator times $\min\{1, (\alpha - r^{\text{upper}}(A))/D(A, p)\}$, maximizing over launch alternatives. The factor is one when $D(A, p) = 0$. It describes an approximate risk-limited insertion opportunity, not simultaneous feasibility of many templates. Floating-point feature calculations do not certify actions; admission still uses the joint oracle. A matched-dimensional control uses the original pattern features with the same risk-remaining and accepted-count context. Both networks have 12,425 parameters.

The first development comparison uses seed 11, 200 sampled training days and validation selection every 40 episodes. Action-dependent random-number consumption produces different sampled training trajectories, with 10,898 versus 10,720 environment transitions and 2,709 versus 2,665 gradient updates. The comparison is not an exactly paired exogenous-day experiment. Resource prices are selected from eight values and the conditional three-training-path LP weight from $\{0.25, 0.5, 1\}$. All current actions face the same joint-risk filter with $\eta = \alpha = 0.01$. Future LPs relax integrality and omit future joint-risk constraints.

Table 6: Development validation under the synthetic joint-risk constraint. One new training seed; no independent confirmation.

Policy	Saving/day	Mean decision ms
Greedy	75.05	0.176
Tuned resource price	79.90	0.211
Pattern TD with risk context	80.03	0.566
Risk-insertion TD	79.99	0.933
Conditional training-path LP	82.17	4.803

All stored final policy-day audits satisfy the finite-law risk bound; the largest bound is approximately 0.008358 and no oracle state limit occurs. Both learned variants select their 40-episode checkpoint and fail to improve consistently with further training. The proposed feature

is approximately 2.66% below the strongest listed baseline and offers no observed advantage over the pattern control. This version is therefore not advanced as a successful method contribution. The saving objective also omits tail-conflict recourse costs and remains a gross simulated work proxy rather than realized operating profit. Fixed exogenous training sequences, value-ranking diagnostics, stronger adapted methods, disturbance calibration and independent confirmation remain necessary.

A subsequent development diagnostic evaluates frozen continuation-value multipliers

$$\{0, 0.25, 0.5, 0.75, 1, 1.5, 2\}.$$

Zero reproduces greedy dispatch on every validation day and one reproduces the stored learned policies. The best grid values are 81.13 for the pattern control and 80.34 for risk-insertion TD, both at multiplier two and both below the conditional LP. This is additional post hoc validation tuning, not independent improvement or evidence that the boundary grid point is globally optimal. Simple rescaling does not close the observed baseline gap.

An independent review reproduced both stored checkpoints on all 33 validation days but identified a protocol-coverage omission: the historical training loop did not independently audit each training-day completion and discarded oracle-limit counts during checkpoint-selection validation. Final stored-policy feasibility is supported; the original files cannot establish that every training query was resolved. A separate audited entry point now records all these episodes without changing the frozen source identity; supplementary runs are identified separately. This distinction is necessary to avoid presenting newly produced audit logs as evidence saved during the original execution.

Chance-constrained UAM occupancy control already appears in Pang et al. [12]; its full formulation has not yet been inspected here. Similarly, conference evidence documents partially joint chance-constrained UAV parcel assignment. These references preclude claiming that chance constraints or action filtering alone establish novelty. The current contribution status remains unconfirmed.

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